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Integrated assembly of an electric motor, hydraulic pump, and electronic drive device and associated cooling configuration

Abstract

An example assembly includes: a main housing (**108**) having an internal chamber (**110**) therein; an electric motor (**102**) disposed in the internal chamber of the main housing and comprising a motor rotor (**114**); a cooling inner ring (**700**) disposed in the internal chamber of the main housing about the electric motor, wherein the cooling inner ring comprises one or more motor cooling fluid channels configured to receive cooling fluid from an external source of cooling fluid and allow cooling fluid to flow about the electric motor to cool the electric motor; a hydraulic pump (**104**) positioned in the main housing, at least partially within the motor rotor of the electric motor; and an enclosure (**173**) coupled to the main housing and comprising (i) an inverter board (**202**) disposed therein, and (ii) one or more inverter cooling fluid channels configured to allow cooling fluid from the external source to cool the inverter board.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION (1) This application is a U.S. National Phase Application pursuant to 35 U.S.C § 371 of International Application No. PCT/US2022/037849 filed on Jul. 21, 2022, which claims priority to U.S. Provisional Application No. 63/297,839, filed Jan. 10, 2022, the entire contents of all of which are herein incorporated by reference.

BACKGROUND
(1) An electric motor can be used to drive a hydraulic pump. For example, an output shaft coupled to a rotor of the electric motor can be coupled to a shaft of the hydraulic pump, such that rotation of the rotor can cause a rotating group of the hydraulic pump to rotate and provide fluid flow.
(2) An electronic drive device including an inverter and motor controller is typically separate from the motor and is connected to wire windings of a stator of the electric motor via cables. It may be desirable to have an assembly that integrates the hydraulic pump and the electronic drive device with the electric motor. This way, mechanical components, such as shafts, bearing, etc., can be shared between hydraulic pump and the motor. It may also be desirable to properly support components of the electric motor and the hydraulic pump and maintain them lubricated to elongate the life of the assembly.
(3) Typically, a hydraulic system may have a system controller that controls the actuator in addition to the controller of the electric motor and hydraulic pump. Thus, multiple controllers may be in communication with each other with cables and signals connecting them. It may thus be desirable to configure the electronic drive device of the assembly to receive actuator sensor information associated with an actuator driven by the hydraulic pump, and the electronic drive device can then control the electric motor and hydraulic pump to achieve commanded motion of the actuator. This way, the system controller can be integrated with the electric motor controller into a single controller and avoid having cables, buses, or signals between separate controllers.
(4) During operation, heat is generated and may cause damage to components of the assembly. It may be desirable to cool components of the electric motor. It may further be desirable to cool the electronic drive device (e.g., the inverter) as it may generate a large amount of heat during operation.
(5) It is with respect to these and other considerations that the disclosure made herein is presented.
SUMMARY
(6) The present disclosure describes implementations that relate to systems an integrated assembly of an electric motor, hydraulic pump, and electronic drive device and associated cooling configuration.
(7) In a first example implementation, the present disclosure describes an assembly comprising: a main housing having an internal chamber therein: an electric motor disposed in the internal chamber of the main housing and comprising a motor rotor: a cooling inner ring disposed in the internal chamber of the main housing about the electric motor, wherein the cooling inner ring comprises one or more motor cooling fluid channels configured to receive cooling fluid from an external source of cooling fluid and allow cooling fluid to flow about the electric motor to cool the electric motor; a hydraulic pump positioned in the main housing, at least partially within the motor rotor of the electric motor; and an enclosure coupled to the main housing and comprising (i) an inverter board disposed therein, and (ii) one or more inverter cooling fluid channels configured to

allow cooling fluid from the external source to cool the inverter board.
(8) In a second example implementation, the present disclosure describes a hydraulic system comprising: an actuator having a first chamber and a second chamber; and a valve assembly configured to control fluid flow to and from the first chamber and the second chamber of the actuator; and the assembly of the first example implementation.
(9) The foregoing summary is illustrative only and is not intended to be in any way limiting. In addition to the illustrative aspects, implementations, and features described above, further aspects, implementations, and features will become apparent by reference to the figures and the following detailed description.

Description

BRIEF DESCRIPTION OF THE FIGURES
(1) The novel features believed characteristic of the illustrative examples are set forth in the appended claims. The illustrative examples, however, as well as a preferred mode of use, further objectives and descriptions thereof, will best be understood by reference to the following detailed description of an illustrative example of the present disclosure when read in conjunction with the accompanying Figures.
(2) FIG. **1** illustrates a perspective view of an assembly, in accordance with an example implementation.
(3) FIG. **2** illustrates a cross-sectional side view of the assembly of FIG. **1**, in accordance with an example implementation.
(4) FIG. **3** illustrates a perspective exploded view of the assembly of FIG. **1**, in accordance with another example implementation.
(5) FIG. **4**A illustrates the cross-sectional view of FIG. **2**, in accordance with an example implementation.
(6) FIG. **4**B illustrates a detailed sectional view showing a pump drive shaft supporting a motor rotor, in accordance with an example implementation.
(7) FIG. **5** illustrates a perspective view of the assembly of FIG. **1** depicting cooling fluid paths, in accordance with an example implementation.
(8) FIG. **6**A illustrates a partial perspective view of the assembly of FIG. **1** showing an inverter cover and a back plate, in accordance with an example implementation.
(9) FIG. **6**B illustrates a partial perspective exploded view of the assembly of FIG. **1** showing the inverter cover and the back plate, in accordance with an example implementation.
(10) FIG. **7**A illustrates a partial perspective view of the assembly of FIG. **1** showing a main housing and a cooling inner ring disposed therein, in accordance with an example implementation.
(11) FIG. **7**B illustrates a partial perspective exploded view of the assembly of FIG. **1** showing the main housing and the cooling inner ring, in accordance with an example implementation.
(12) FIG. **8** illustrates a perspective view of another cooling inner ring, in accordance with an example implementation.
(13) FIG. **9** illustrates a perspective view of another cooling inner ring, in accordance with an example implementation.
(14) FIG. **10** illustrates a perspective view of another cooling inner ring, in accordance with an example implementation.
(15) FIG. **11** illustrates a hydraulic system including the assembly of FIG. **1** operating in an open-circuit configuration, in accordance with an example implementation.
(16) FIG. **12** illustrates a hydraulic system having the assembly of FIG. **1** operating in a closed-circuit configuration, in accordance with an example implementation.
DETAILED DESCRIPTION

(17) The present disclosure relates to integrating or embedding a hydraulic pump and an electronic drive device (e.g., including a motor controller and an inverter) with an electric motor to have an assembly providing a compact configuration that reduces cost by sharing components, saves space, and enhances reliability. The assembly also includes a cooling configuration where an external source provides cooling fluid to cool the electronic drive device and the electric motor.

(18) The rotor of the electric motor is properly supported at two points, on its exterior surface and interior surface, to provide enhanced support while allowing components of the hydraulic pump to be disposed, at least partially, within the rotor. The drive shaft of the hydraulic pump is drivingly connected to the rotor of the electric motor via a spline connection, and the spline connection is maintained lubricated and sealed to enhance life and performance of the assembly.

(19) FIG. **1** illustrates a perspective view of an assembly **100**, FIG. **2** illustrates a cross-sectional side view of the assembly **100**, and FIG. **3** illustrates a perspective exploded view of the assembly **100**, in accordance with an example implementation. FIGS. **1-3** are described together.

(20) The assembly **100** comprises an electric motor **102**, a hydraulic pump **104**, and an electronic drive device **106** integrated together. The assembly **100** includes a main housing **108** having an internal chamber **110** therein in which components of the electric motor **102** and the hydraulic pump **104** are disposed.

(21) The electric motor **102** includes a stator **112** fixedly-positioned within the internal chamber **110** of the main housing **108**. The stator **112** is configured to generate a magnetic field. Particularly, the stator **112** can include wire windings (not shown) wrapped about a body (e.g., a lamination stack) of the stator **112**, and when electric current is provided through the wire windings, a magnetic field is generated.

(22) The electric motor **102** further includes a motor rotor **114** positioned within the stator **112**. The electric motor **102** can further include magnets **116** mounted to the motor rotor **114** in an annular space between the stator **112** and the motor rotor **114**. In an example, the magnets **116** include two circumferential arrays of magnets axially-spaced and disposed about the motor rotor **114** as shown in FIG. **3**.

(23) The magnets **116** are configured to interact with the magnetic field generated by the stator **112** in order to rotate the motor rotor **114** and produce torque. In other example implementation, a different type of electric motor might be used that does not include permanent magnets.

(24) The hydraulic pump **104** is mounted within the main housing **108** and, at least partially, within the motor rotor **114** and the stator **112** of the electric motor **102**. The assembly **100** has a pump port block **117** having an inlet port **118** and an outlet port **120**. The pump port block **117** is coupled to the main housing **108** via a plurality of fasteners or bolts such as bolt **121** (see FIGS. **1** and **3**).

(25) The hydraulic pump **104** can be configured, for example, as a vane pump. Particularly, the hydraulic pump **104** has a pump cover **124** and a vane cartridge **126**. The vane cartridge **126** has a pump housing **128**, a pump rotor **130**, vanes such as vane **131** and vane **132**, and a pump drive shaft **134**.

(26) The motor rotor **114** has a cylindrical portion **136** and a spindle portion **138**. The spindle portion **138** is supported within an electronic device housing **140** of the electronic drive device **106** via a bearing **142** disposed about an exterior surface of the spindle portion **138** of the motor rotor **114** to allow the motor rotor **114** to rotate relative to the main housing **108** and the electronic device housing **140**. Further, the pump drive shaft **134** provides additional support for the motor rotor **114**.

(27) FIG. **4**A illustrates the cross-sectional view of FIG. **2**, and FIG. **4**B illustrates a detailed sectional view showing the pump drive shaft **134** supporting the motor rotor **114**, in accordance with an example implementation. Particularly, FIG. **4**B is a zoomed-in view of the cross-sectional view of FIG. **4**A illustrating the interface between the pump drive shaft **134** and the spindle portion **138** of the motor rotor **114**.

(28) The pump drive shaft **134** is rotatably-coupled to the motor rotor **114**. Referring to FIGS. **3** and **4**B together, the pump drive shaft **134** has splines **144** formed about the exterior surface of the

pump drive shaft **134** and configured to engage with respective splines **146** formed on an interior surface of the spindle portion **138** of the motor rotor **114**. This way, the motor rotor **114** is drivingly connected to the pump drive shaft **134** and is configured to transmit rotary motion to the pump drive shaft **134** during operation of the assembly **100**.

(29) As mentioned above, the motor rotor **114**, and particularly the spindle portion **138** thereof, is supported about its exterior surface via the bearing **142**. Additionally, the pump drive shaft **134** has an increased diameter portion **147** and shoulder **148** that provide support for the interior surface of the spindle portion **138** of the motor rotor **114** as shown in FIG. **4**B. As such, the motor rotor **114** is supported at two areas or points: an area about its exterior surfaces supported by the bearing **142** and an area about its interior surface where the motor rotor **114** is supported by the pump drive shaft **134**. With this configuration, the pump drive shaft **134** provides an extended support for the motor rotor **114** and may preclude misalignments or twisting of the motor rotor **114** during operation.

(30) During operation, as the motor rotor **114** rotates, the spline engagement with the pump drive shaft **134** causes the rotary motion of the motor rotor **114** to be transmitted to the pump drive shaft **134**. It may be desirable to maintain the spline engagement between the splines **146** and the splines **144** lubricated. Such lubrication elongates the life of the components (e.g., the motor rotor **114** and the pump drive shaft **134**) of the assembly **100**. For example, grease may be used to lubricate the spline engagement between the splines **146** and the splines **144**.

(31) It may be desirable to maintain the grease or any lubrication fluid at the area of spline engagement. As such, the assembly **100** includes a first seal **150** (e.g., an O-ring) disposed in a groove formed about an exterior surface of the pump drive shaft **134**. Further, the assembly **100** includes a cap **152** inserted within the spindle portion **138** of the motor rotor **114**, and the cap **152** has a second seal **154** (e.g., an O-ring) disposed in a groove formed about a respective exterior surface of the cap **152**.

(32) The first seal **150** and the second seal **154** straddle the splines **144**, **146**. This way, any lubricant at the area of spline engagement between the splines **146** and the splines **144** is sealed and remains between the first seal **150** and the second seal **154**. With this configuration, the lubricant might not leak outside the area of spline engagement.

(33) Referring back to FIGS. **2-3**, the assembly **100** further includes another bearing **156** disposed about the exterior surface of, and provides support to, the pump drive shaft **134**. Particularly, the bearing **156** is disposed between the exterior surface of the pump drive shaft **134** and the interior surface of the pump cover **124** and allows the pump drive shaft **134** to rotate relative to the pump cover **124**. The assembly further includes a shaft seal **158** also disposed between the exterior surface of the pump drive shaft **134** and the interior surface of the pump cover **124** and configured to preclude leakage of fluid from the hydraulic pump **104** into the electric motor **102**.

(34) Referring to FIG. **2**, fluid received at the inlet port **118** flows through a first feeding passage **160** and a second feeding passage **162**. Fluid flows from the first feeding passage **160** to a pump chamber **164**. Fluid also flows from the second feeding passage **162**, through arcuate fluid passage **165** and arcuate fluid passage **166** formed in the vane cartridge **126** to join fluid from the first feeding passage **160** at the pump chamber **164**.

(35) Having two feeding passages, rather than one, may advantageously increase the feeding capacity to the hydraulic pump **104**. As more fluid is being fed to the hydraulic pump **104**, the likelihood of occurrence of cavitation is decreased or eliminated.

(36) The pump rotor **130** is configured to be eccentrically supported within the pump housing **128**, which may have a cycloidal interior surface. The pump rotor **130** is located close to the interior surface wall of the pump housing **128** such that a crescent-shaped cavity is formed therebetween. The vanes, such as the vanes **131**, **132** fit within slots in the pump rotor **130**.

(37) As the pump drive shaft **134** rotates (when the motor rotor **114** rotates), the pump rotor **130** also rotates. As the pump rotor **130** rotates, centrifugal force, hydraulic pressure, pushrods, and/or

springs push the vanes **131**, **132** radially-outward toward the interior surface of the pump housing **128**. Fluid is then forced or sucked from the pump chamber **164** into the pump housing **128** via holes such as hole **167** shown in FIG. **2**. Particularly, fluid enters pockets created by the vanes **131**, **132**, the pump rotor **130**, and the pump housing **128**.

(38) As the pump rotor **130** continues rotating, the vanes **131**, **132** sweep the fluid to the opposite side of the crescent-shaped cavity where it is squeezed through discharge holes in the pump housing **128**. Fluid is then forced to the outlet port **120** from which where fluid exits the assembly **100**. Fluid is then provided to an actuator fluidly-coupled to the assembly **100** as described below with respect to FIGS. **10-11**.

(39) A vane pump is used herein as an example for illustration. It should be understood that other types of pumps such as a gear pump or a piston pump can be used as well.

(40) Referring to FIG. **3**, the electronic device housing **140** is coupled to the main housing via bolts such as bolt **168**. The electronic device housing **140** is further coupled to an electronics housing cover **170** via a plurality of bolts, such as bolt **172**. With this configuration, the electronic device housing **140** and the electronics housing cover **170** form an enclosure **173** in which electronic boards and components of the electronic drive device **106** are disposed. This way, the electronic drive device **106** is integrated with the electric motor **102** and the hydraulic pump **104** in the assembly **100**.

(41) The electronic drive device **106** can include one or more electronic boards such as a controller board **200** and an inverter board **202** that are electrically-coupled and axially-offset from each other as depicted. The controller board **200** and the inverter board **202** can be configured as printed circuit boards (PCBs). A PCB mechanically supports and electrically connects electronic components (e.g., microprocessors, integrated chips, capacitors, resistors, etc.) using conductive tracks, pads, and other features etched from one or more sheet layers of copper laminate onto and/or between sheet layers of a nonconductive substrate. Components are generally soldered onto the PCB to both electrically connect and mechanically fasten them to it.

(42) The inverter board **202** can be separated from, and coupled to, the controller board **200** via stand-offs such as stand-off **204** shown in FIG. **3**. The inverter board **202** can include a plurality of bus bars that are electrically-conductive and are configured to receive direct current (DC) power and provide the power to components mounted to the inverter board **202**.

(43) As an example, the DC power can be provided to the inverter board **202** from a battery. With this configuration, DC power is provided to the bus bars, which then transmit the power to other components of the inverter board **202**.

(44) The inverter board **202** can be configured as a power converter that converts DC power received at the inverter board **202** to three-phase, alternating current (AC) power that can be provided to wire windings of the stator **112** to drive the electric motor **102**. For example, the inverter board **202** can include a semiconductor switching matrix mounted to the inverter board **202** and configured to be electrically-connected to a positive DC terminal and to a negative DC terminal. The inverter board **202** can further include a plurality of capacitors disposed in the axial space between the inverter board **202** and the controller board **200**.

(45) The semiconductor switching matrix can include any arrangement of semiconductor switching devices that supports DC to three-phase power conversion. For example, the semiconductor switching matrix can include a three-phase, with bridge elements electrically-coupled to input DC terminals and connected to three-phase AC output terminals.

(46) In an example, the semiconductor switching matrix includes a plurality of transistors (e.g., an Insulated Gate Bipolar Transistor or a metal-oxide-semiconductor field-effect transistor). The transistors are switchable between an activated or “on” state and a deactivated or “off” state, e.g., via a pulse width modulated (PWM) signal provided by a microprocessor mounted to the controller board **200**. A microprocessor can comprise one or more processors. A processor can include a general purpose processor (e.g., an INTEL® single core microprocessor or an INTEL® multicore

microprocessor), or a special purpose processor (e.g., a digital signal processor, a graphics processor, or an application specific integrated circuit (ASIC) processor). A processor can be configured to execute computer-readable program instructions (CRPI) to perform the operations described throughout herein. A processor can be configured to execute hard-coded functionality in addition to or as an alternative to software-coded functionality (e.g., via CRPI).
(47) As the transistors of the semiconductor switching matrix are activated and deactivated at particular times via the PWM signal, AC voltage waveforms are generated at the AC output terminals. As such, the voltage waveforms at the AC output terminals are pulse width modulated and swing between voltage potential DC+ and voltage potential DC−. The AC voltage waveforms are then provided to the wire windings of the stator **112** to drive the electric motor **102**. The controller board **200** and the inverter board **202** can include several junctions to facilitate receiving and transmitting signals and power therebetween and to other components of the assembly **100** or external components.
(48) The electronic drive device **106** can include a plurality of sensors. For example, the electronic drive device **106** can include temperature sensors configured to provide information indicative of an operating temperature of the electric motor **102** and/or the fluid temperature of hydraulic fluid flowing through the hydraulic pump **104**. Another temperature sensor may be mounted to the inverter board **202** to indicate a temperature of the inverter board **202**. The electronic drive device **106** can also include Hall-Effect current sensors that provide information indicative of electric current level in windings of the stator **112**.
(49) The electronic drive device **106** can further include a pressure sensor indicative of pressure level of fluid within the assembly **100**. The electronic drive device **106** can also include a rotary position sensor configured to provide sensor information indicative of angular position of the motor rotor **114** and the pump drive shaft **134**. The rotary position sensor information can be used by the microprocessor controlling the electric motor **102** so as to control speed and torque produced by the motor rotor **114** in a closed-loop feedback control configuration.
(50) For example, the controller board **200** can include a sensor chip or encoder **206** to be mounted near the spindle portion **138** of the motor rotor **114** as show in FIG. **2**. The encoder **206** can be configured to interact with a magnet **208** disposed in the cap **152**. The encoder **206** is configured as an electro-mechanical device that converts the angular position or motion of the motor rotor **114** to analog or digital output signals that are provided to the microprocessor controlling the electric motor **102**. In an example implementation, to enhance accuracy of the sensor information provided by the encoder **206**, the magnetic polarity of the magnet **208** is aligned with, or have the same orientation as, respective magnetic polarity of respective magnets (i.e., the magnets **116**) of the electric motor **102**.
(51) As shown in FIGS. **1-3**, the electronic drive device **106** can further include an electric connector **210** configured as a hollow plastic component housing a plurality of conductor pins that are electrically-connected to the conductive tracks of the controller board **200**. A connector socket (not shown) having female pins can be mounted to or inserted into the electric connector **210** such that the conductor pins contact the female pins of the connector socket. Wires can be connected to the female pins so as to provide signals to, and receive signals from, the conductor pins of the electric connector **210**.
(52) With this configuration, the electronic drive device **106** can receive via the electric connector **210** various inputs and sensor signals and provide commands in response to the received information. For example, the electronic drive device **106** can receive a command signal from a central controller or from an input device of a machine (e.g., a joystick of a hydraulic machine such as a wheel loader, backhoe, or excavator) indicative of a desired fluid pressure and fluid flow rate to be provided by the hydraulic pump **104**. The electronic drive device **106** can then control the AC power provided to the stator **112** to generate a particular speed and torque at the pump drive shaft **134** to provide the desired fluid pressure level and flow rate. The electronic drive device **106** can

also provide sensor signals (e.g., from the encoder **206**) to another central controller via the electric connector **210**.
(53) As such, the electronic drive device **106** can be used as a controller for the assembly **100** as well as the actuator controlled by the assembly **100**. Particularly, the electronic drive device **106** receives command inputs and sensor signals from sensors internal to the assembly **100** and sensors associated with the actuator, and responsively controls the electric motor **102** and the hydraulic pump **104** to achieve desired or commanded motion of the actuator.
(54) During operation of the assembly **100**, heat is generated by the electric motor **102** and the heat is distributed throughout multiple components within the electric motor **102**. For example, heat is generated due to losses within the stator slot-windings, stator end-windings, stator laminations, rotor laminations, and rotor magnets or conductors. The distribution of the generated heat within the components is dependent on the motor type and the operating condition (torque/speed) of the motor. The heat generated within the electric motor **102** can raise the internal temperature, making the coil of the electric motor **102** more susceptible to burn out. It may thus be desirable to cool the electric motor **102** to preclude the temperature from exceeding a threshold safe temperature.
(55) Further, during operation, as the inverter board **202** converts DC power to AC power, it generates heat. This heat is added to the ambient temperature of the enclosure **173** formed by the electronic device housing **140** and the electronics housing cover **170**. The hotter the inverter board **202** and the enclosure **173** become, the less AC power the inverter board **202** delivers. As such, it may be desirable to also cool the enclosure **173** and the inverter board **202**. The assembly **100** is configured to receive cooling fluid from an external source and route the cooling fluid to both the enclosure **173** (e.g., the electronics housing cover **170**) to cool the inverter board **202** and to the main housing **108** to cool the electric motor **102**.
(56) FIG. **5** illustrates a perspective view of the assembly **100** depicting cooling fluid paths, in accordance with an example implementation. The assembly **100** is configured to receive cooling fluid from an external source **500** of cooling fluid.
(57) In an example, the external source **500** can be a pump that draws cooling fluid such as water glycol from a fluid reservoir and displaces the cooling fluid to the assembly **100**. In another example, the external source **500** can be another hydraulic circuit (e.g., see boost circuit **404** in FIG. **12** described below) included in the hydraulic system comprising the assembly **100**.
(58) In another example, as discussed below with respect to FIG. **11**, the external source **500** can be a hydraulic actuator (e.g., a hydraulic cylinder or hydraulic motor) included in the hydraulic system. In this example, a portion of the fluid discharged from the hydraulic actuator is routed to the assembly **100** for cooling while a remaining amount of fluid discharged from the hydraulic actuator flows to a fluid reservoir (e.g., a fluid tank).
(59) Cooling fluid from the external source **500** is provided through a fluid line **502** (e.g., a hose, tube, pipe, etc.) to a junction **504**. At the junction **504**, the cooling fluid is divided or branched such that a portion of the cooling fluid branches out to flow through a fluid line **506** to a motor cooling fluid inlet port **508**, while another portion flows through a flow control valve **510** to an inverter cooling fluid inlet port **512**.
(60) The flow control valve **510** is configured to apportion the cooling fluid between the electric motor **102** and the electronic drive device **106**. For example, a controller (e.g., a microprocessor of the controller board **200**) can be configured to receive sensor information indicative of temperature of the electric motor **102** and the temperature of the inverter board **202**. Based on the sensor information, the controller determines whether to provide more cooling fluid to the electric motor **102** or the electronic drive device **106**. Based on the determination, the controller provides a signal to a solenoid **514**, for example, to operate the flow control valve **510** so as to increase or decrease fluid resistance in the path of fluid provided to the inverter cooling fluid inlet port **512**. This way, cooling fluid can be apportioned between the electric motor **102** and the electronic drive device **106** based on cooling requirements.

(61) In an example, the flow control valve **510** can be manually-operated. In another example, the assembly **100** might not include the flow control valve **510**.
(62) Fluid provided to the inverter cooling fluid inlet port **512** circulates within the enclosure **173** (e.g., through the electronics housing cover **170**), thereby absorbing heat generated by the inverter board **202**, and is then provided to an inverter cooling fluid outlet port **516** from which cooling fluid is discharged to fluid line **518**. Similarly, fluid provided to the motor cooling fluid inlet port **508** circulates around the electric motor **102** (e.g., about the main housing **108**), thereby absorbing heat generated by the electric motor **102**, and is then provided to a motor cooling fluid outlet port **520** from which cooling fluid is discharged to fluid line **522**.
(63) Fluid from the fluid line **522** merges with fluid from the fluid line **518** at junction **524**, and fluid is then provided to fluid line **526**, which can be fluidly-coupled to a fluid reservoir. The fluid reservoir can be a fluid reservoir dedicated to the cooling fluid or can be a fluid reservoir of hydraulic fluid of the hydraulic system.
(64) The several fluid lines depicted in FIG. **5** can include multiple fittings as appropriate. Further, although the fluid lines or paths of the cooling fluid and other components such as the flow control valve **510** are depicted as being external to the assembly **100**, it is contemplated that at least some of the fluid paths or components can be embedded within the assembly **100**.
(65) FIG. **6**A illustrates a partial perspective view of the assembly **100** showing the electronics housing cover **170** and a back plate **600**, and FIG. **6**B illustrates a partial perspective exploded view of the assembly **100** showing the electronics housing cover **170** and the back plate **600**, in accordance with an example implementation. The back plate **600** can be coupled to the electronics housing cover **170** via a plurality of fasteners such as fastener **601**.
(66) The enclosure **173** is configured to have one or more inverter cooling fluid channels to cool the inverter board **202**. For example, fluid provided from the external source **500** shown in FIG. **5** is provided to the inverter cooling fluid inlet port **512** (via the flow control valve **510**), and fluid is then provided to an inverter cooling fluid channel **602** formed in the electronics housing cover **170** and the back plate **600**. For example, fluid received at the inverter cooling fluid inlet port **512** can be provided to a first end **604** of the inverter cooling fluid channel **602** (e.g., the inlet opening in the back plate **600** is aligned with the first end **604**). The cooling fluid then traverses the inverter cooling fluid channel **602** until it reaches a second end **606** of the inverter cooling fluid channel **602**. The second end **606** can be aligned with an outlet opening in the back plate **600**, where the outlet opening provides fluid to the inverter cooling fluid outlet port **516**.
(67) As depicted, the inverter cooling fluid channel **602** zigzags or loops through the electronics housing cover **170**. This way, the surface area through which heat is dissipated is increased.
(68) In an example, the inverter cooling fluid channel **602** is formed in the electronics housing cover **170** only. In another example, a portion of the inverter cooling fluid channel **602** is formed in the electronics housing cover **170** and another mating portion of the inverter cooling fluid channel **602** is formed in the back plate **600**. For example, as depicted in FIG. **2**, if the inverter cooling fluid channel **602** comprises a cylindrical fluid passage or conduit, half the conduit can be formed in the electronics housing cover **170** and the other mating half is formed in the back plate **600**.
(69) The electronics housing cover **170** can further include a seal groove **608** in which a seal (e.g., an O-ring) can be disposed. Such seal is configured to seal the cooling fluid flowing through the inverter cooling fluid channel **602** and precludes leakage of cooling fluid to an external environment of the assembly **100**, i.e., fluid is contained at the interface between the electronics housing cover **170** and the back plate **600**.
(70) While the inverter cooling fluid channel **602** is shown in FIGS. **6**A-**6**B as being formed in the electronics housing cover **170** and the back plate **600**, other configurations are possible. The cooling channel can be formed anywhere through the enclosure **173** (i.e., can be formed in one or more of the back plate **600**, the electronics housing cover **170**, or the electronic device housing **140**) such that the cooling fluid circulates about the inverter board **202** and absorbs heat generated

therefrom.
(71) FIG. **7**A illustrates a partial perspective view of the assembly **100** showing the main housing **108** and a cooling inner ring **700** disposed therein, and FIG. **7**B illustrates a partial perspective exploded view of the assembly **100** showing the main housing **108** and the cooling inner ring **700**, in accordance with an example implementation. The cooling inner ring **700** is retained within the main housing **108** via retaining pins such as retaining pin **702**, which is configured to extend into a hole made in the cooling inner ring **700** (see FIG. **2**) to retain it to the main housing **108**.
(72) The cooling inner ring **700** is configured to having one or more motor cooling fluid channels providing cooling paths for the cooling fluid received at the motor cooling fluid inlet port **508** of the main housing **108**. Particularly, as shown in FIG. **7**B, the cooling inner ring **700** can have a plurality of arcuate protrusions or arcuate ridges such as arcuate ridge **704**, arcuate ridge **706**, arcuate ridge **708**, and arcuate ridge **710**. The cooling inner ring **700** can also have another set of arcuate ridges opposite the arcuate ridges **704**-**710**, as shown in FIG. **7**B, such that a motor cooling fluid channel **712** is formed therebetween.
(73) Cooling fluid received via the motor cooling fluid inlet port **508** is provided to the cooling inner ring **700** at an inlet end **714** of the motor cooling fluid channel **712**. Cooling fluid has several path to traverse thereafter under pressure from the external source **500** providing the cooling fluid. For example, cooling fluid can traverse the motor cooling fluid channel **712** to an outlet end **716** of the motor cooling fluid channel **712**.
(74) Further, cooling fluid can traverse an exterior surface of the cooling inner ring **700** through arcuate motor cooling fluid channels formed between the arcuate ridges **704**-**710**, for example. Such arcuate motor cooling fluid channels form parallel paths for the cooling fluid to traverse while absorbing heat throughout the surfaces of the arcuate ridges **704**-**710**.
(75) Under pressure from the external source **500**, cooling fluid traverses the various paths or channels of the cooling inner ring **700** and reaches the outlet end **716**, which is aligned with the motor cooling fluid outlet port **520**. Fluid is then discharged from the motor cooling fluid outlet port **520**, and flows through the fluid line **522** to merge with the cooling fluid that has traversed the electronics housing cover **170** as described above with respect to FIG. **5**.
(76) Although FIG. **7**B illustrates the cooling inner ring **700** having parallel paths (e.g., arcuate channels between the arcuate ridges **704**-**710**), the cooling inner ring **700** can be configured differently. For instance, another cooling inner ring can have a serial path (e.g., one spiral or zigzagging motor cooling fluid channel). Another cooling inner ring may have protrusion of various shapes to increase the surface area through which heat is dissipated.
(77) FIG. **8** illustrates a perspective view of another cooling inner ring **800**, in accordance with an example implementation. The cooling inner ring **800** has a spiral motor cooling fluid channel **802** that is formed as a looping depression about the exterior surface of the cooling inner ring **800**. The spiral motor cooling fluid channel **802** has an inlet end **804** aligned with the motor cooling fluid inlet port **508** to receive cooling fluid therefrom. Cooling fluid then circulates about the exterior surface of the cooling inner ring **800**, and the side surfaces of the spiral motor cooling fluid channel **802** may increase the surface area through which heat is dissipated to the cooling fluid.
(78) Fluid then reaches an outlet end **806**, which is aligned with the motor cooling fluid outlet port **520**. Fluid is then discharged from the motor cooling fluid outlet port **520**, and flows through the fluid line **522** to merge with the cooling fluid that has traversed the electronics housing cover **170** as described above with respect to FIG. **5**.
(79) FIG. **9** illustrates a perspective view of another cooling inner ring **900**, in accordance with an example implementation. The cooling inner ring **900** has a plurality protrusions **902** having a polygonal shape (e.g., closed plane shape bounded by straight sides such a quadrilateral, a pentagon, a hexagon, etc.).
(80) The protrusions **902** are interposed longitudinally (i.e., along a longitudinal axis of the cooling inner ring **900**) between an inlet circular channel **904** and an outlet circular channel **906**. The inlet

circular channel **904** receives cooling fluid from the motor cooling fluid inlet port **508**, and cooling fluid is discharged from the outlet circular channel **906** to the motor cooling fluid outlet port **520**.
(81) Due to the configuration of the protrusions **902** being polygonal in shape, various cooling paths are formed therebetween. Fluid received at the motor cooling fluid inlet port **508** is provided to the inlet circular channel **904**, and the cooling fluid is then diffused throughout the various motor cooling fluid channels/paths formed between the protrusions **902**. The multiple raised sides of each protrusion may increase the surface area through which heat is dissipated and absorbed by the cooling fluid.
(82) Under pressure from the external source **500**, fluid is diffused through the protrusions **902** until it reaches the outlet circular channel **906**, which is aligned with the motor cooling fluid outlet port **520**. Fluid is then discharged from the motor cooling fluid outlet port **520**, and flows through the fluid line **522** to merge with the cooling fluid that has traversed the electronics housing cover **170** as described above with respect to FIG. **5**.
(83) FIG. **10** illustrates a perspective view of another cooling inner ring **1000**, in accordance with an example implementation. The cooling inner ring **1000** has a zigzagging motor cooling fluid channel **1002** that is formed as a depression that zigzags longitudinally as it traverses an exterior surface of the cooling inner ring **1000**. The zigzagging motor cooling fluid channel **1002** has an inlet end **1004** aligned with the motor cooling fluid inlet port **508** to receive cooling fluid therefrom.
(84) Under pressure from the external source **500**, cooling fluid circulates and zigzags about the exterior surface of the cooling inner ring **1000** as it traverses the zigzagging motor cooling fluid channel **1002**. The side surfaces of the zigzagging motor cooling fluid channel **1002** may increase the surface area through which heat is dissipated to the cooling fluid.
(85) Fluid then reaches an outlet end **1006**, which is aligned with the motor cooling fluid outlet port **520**. Fluid is then discharged from the motor cooling fluid outlet port **520**, and flows through the fluid line **522** to merge with the cooling fluid that has traversed the electronics housing cover **170** as described above with respect to FIG. **5**.
(86) Referring back to FIG. **2**, the assembly **100** is depicted with the cooling inner ring **700**. However, it should be understood that the cooling inner rings **800**, **900**, **1000** could be used.
(87) Further, in other example implementations, the cooling channels may be formed within the main housing **108**. In these examples, the cooling inner ring might not be used; rather, the main housing **108** operates as a cooling jacket with cooling channels formed therethrough. Any other configuration in which cooling fluid is allowed to circulate through a cooling fluid channel about the electric motor **102** to absorb heat generated therefrom can be used.
(88) As shown in FIG. **2**, the cooling inner ring **700** has external annular grooves at both ends, such as annular groove **174** and annular groove **176**. The annular grooves **174**, **176** are configured to receive seals such as O-rings (not shown) to seal the cooling fluid between the main housing **108** and the cooling inner ring **700**.
(89) Further, the main housing **108** is mounted to the electronic device housing **140**, wherein the electronic device housing **140** has a cylindrical protrusion **178**, the exterior surface of which interfaces with an interior surface of the main housing **108**. Another seal **180** disposed in an annular groove formed in the exterior surface of the cylindrical protrusion **178** can further seal the cooling fluid and preclude it from entering the internal chamber **110** in which the electric motor **102** is disposed.
(90) As mentioned above, the external source **500** of cooling fluid can have various configurations. It can be a pump that draws cooling fluid such as water glycol from a fluid reservoir and displaces the cooling fluid to the assembly **100**, or the external source **500** can be another component or circuit of the hydraulic system in which the assembly **100** is used.
(91) FIG. **11** illustrates a hydraulic system **300** including the assembly **100** operating in an open-circuit configuration, in accordance with an example implementation. In FIG. **11**, hydraulic fluid

lines are depicted as solid lines, whereas command and sensor signals are depicted in dashed lines. Also, not all signal lines are shown to reduce visual clutter in the drawing. A power source such a battery or an electric generator is configured to provide direct current power to the assembly **100**. The power source is not shown to reduce visual clutter in the drawing.

(92) The hydraulic system **300** also includes a fluid reservoir **302** of fluid that can store fluid at a low pressure, e.g., 0-70 pounds per square inch (psi). The inlet port **118** of the assembly **100** is fluidly-coupled to the fluid reservoir **302**. The assembly **100** thus receives fluid from the fluid reservoir **302** through the inlet port **118**, then discharges fluid through the outlet port **120** as described above.

(93) The hydraulic system **300** also includes a manifold or valve assembly **304**. The valve assembly **304** can include fluid passages and a plurality of solenoid-actuated valves (e.g., one or more directional control valves, flow control valves, load-holding valves, etc.). The solenoid-actuated valves control fluid flow through such fluid passages.

(94) In an example, the valve assembly **304** is coupled to the assembly **100**. For instance, the valve assembly **304** can be mounted to the pump port block **117**. The valve assembly **304** can thus control fluid flow to and from the inlet port **118** and the outlet port **120**.

(95) The controller board **200** can include electronic valve drivers that control electric current and voltage signals provided to solenoid coils of the solenoid-actuated valves of the valve assembly **304** to control actuation and state of operation of the solenoid valves. This way, the electronic drive device **106** can be configured to control state of operation of the solenoid-actuated valves and fluid flow through the valve assembly **304**.

(96) The valve assembly **304** is configured to direct fluid flow to and from an actuator **306**. The actuator **306** includes a cylinder **308** and a piston **310** slidably accommodated in the cylinder **308**. The piston **310** includes a piston head **312** and a rod **314** extending from the piston head **312** along a central longitudinal axis direction of the cylinder **308**. The piston head **312** divides the inside space of the cylinder **308** into a first chamber **316** and a second chamber **318**.

(97) The electronic drive device **106** of the assembly **100** can receive input information comprising sensor information via signals from various sensors or input devices in the hydraulic system **300**, and in response provide electrical signals to various components of the hydraulic system **300**. For example, the electronic drive device **106** can receive actuator sensor information from a position sensor and/or a velocity sensor coupled to the piston **310** information indicative of the position x and velocity {dot over (x)} of the piston **310**. Additionally or alternatively, the electronic drive device **106** can receive from a pressure sensor **320** coupled to the first chamber **316** and/or a pressure sensor **322** coupled to the second chamber **318** information indicative of pressure level p of fluid in the chambers **316**, **318** or indicative of a magnitude of a load applied to piston **310**.

(98) As mentioned above, the electronic drive device **106** can also receive sensor information from sensors of the assembly **100** (e.g., the encoder **206**) indicative of speed of the motor rotor **114**, pressure level of fluid discharged from the hydraulic pump **104**, temperature of the cooling fluid, temperature of the electric motor **102**, temperature of the inverter board **202**, etc. The electronic drive device **106** can also receive an input (e.g., from a joystick of a machine) indicative of a commanded or desired speed for the piston **310**. The electronic drive device **106** can then provide signals to the electric motor **102** of the assembly **100** and the valve assembly **304** to move the piston **310** at a desired commanded speed in a controlled manner.

(99) For example, to extend the piston **310** (i.e., move the piston **310** up in FIG. **11**), the electronic drive device **106** actuates the electric motor **102**, which causes the hydraulic pump **104** to draw fluid from the fluid reservoir **302** through the inlet port **118**, then provide fluid through the outlet port **120** to the valve assembly **304**. The electronic drive device **106** also sends command signals to one or more valves to operate the valve assembly **304** in a first state. As a result, pressurized fluid provided from the assembly **100** via the outlet port **120** flows through the valve assembly **304** to the first chamber **316**. As the piston **310** extends, fluid forced out of the second chamber **318** flows

to the valve assembly **304**, which directs the fluid via return line **324** to the fluid reservoir **302**. The electronic drive device **106** operates the electric motor **102** at a particular speed and torque to provide a particular amount of flow to the actuator **306** at a particular pressure to move the piston **310** at a desired speed, while controlling pressure level in the first chamber **316** and/or the second chamber **318**.

(100) To retract the piston **310**, the electronic drive device **106** can send command signals to one or more valves to operate the valve assembly **304** in a second state in which the valve assembly **304** directs fluid received from the outlet port **120** to the second chamber **318**. As the piston **310** retracts, fluid in the first chamber **316** is forced out of the first chamber **316** to the valve assembly **304**, which directs the fluid via the return line **324** to the fluid reservoir **302**.

(101) With this configuration, whether the piston **310** is extending or retracting, fluid discharged from the actuator **306** is provided via the return line **324** to the fluid reservoir **302**. At the same time, the fluid discharged from the actuator **306** is also used as the cooling fluid that reduces the temperature of the electric motor **102** and the inverter board **202**.

(102) Particularly, fluid flowing through the return line **324** branches out at junction **326** to cooling fluid line **328**, which is fluidly-coupled to the assembly **100** as shown in FIG. **11** to provide cooling fluid thereto (e.g., via the fluid line **502**). To provide cooling fluid at a particular desired pressure (e.g., 50 psi), the hydraulic system **300** includes a valve **330** configured to allow pressure level upstream of the valve **330** (e.g., at the junction **326**) to be higher than pressure level in the fluid reservoir **302**.

(103) As an example, the valve **330** can be a relief valve with a pressure setting of the desired pressure level. In another example, the valve **330** can include an orifice that restricts fluid flow therethrough, and thus allows pressure level to increase upstream. In another example, the valve **330** can include a spring-loaded check valve. Such a spring-loaded check valve can include a check element (a ball or poppet) that is spring-loaded or biased by a spring to a closed position. When pressure level upstream of the valve **330** reaches a preset pressure level that overcomes the force that the spring applies to the check element, the valve **330** opens and fluid is allowed to flow to the fluid reservoir **302**. This way, fluid provided through the cooling fluid line **328** has the preset pressure level set by the spring. The valve **330** can be a combination of the example configurations mentioned above or any other valve configuration that allows pressure level to build up at the junction **326**.

(104) In another example, the valve **330** can be a proportional electronically-actuated valve. In this example, the controller (e.g., the controller board **200**) can send a signal to an actuator (e.g., a solenoid) of the valve **330** to vary the pressure level upstream of the valve **330** based on cooling requirements (e.g., to provide sufficient cooling in a particular condition). A signal lines is not shown to the valve **330** to reduce visual clutter in the drawing.

(105) The cooling fluid provided via the cooling fluid line **328** to the assembly **100** then circulates through the main housing **108** (see FIGS. **7**A-**10**) and through the electronics housing cover **170** (see FIGS. **6**A-**6**B), and is then provided through the fluid line **526** to cooling fluid return line **332**, which provides the cooling fluid back to the fluid reservoir **302**.

(106) In an example, the electronic drive device **106** of the assembly **100** can receive sensor information indicative of the temperatures of the electric motor **102** and the inverter board **202**. In response, the electronic drive device **106** controls the solenoid **514** of the flow control valve **510** to apportion cooling fluid between the electric motor **102** (e.g., to the cooling inner ring **700**, **800**, **900**, **1000**) and the electronics housing cover **170** such that the respective temperatures of the electric motor **102** and the inverter board **202** remain below safe operating temperatures.

(107) The hydraulic system **300** can be referred to as an open-circuit or open loop system where the assembly **100** draws fluid from the fluid reservoir **302** and then provides fluid to the actuator **306** via the valve assembly **304**, and fluid discharged from the actuator **306** returns to the fluid reservoir **302** via the valve assembly **304**. Alternatively, the assembly **100** can be used in a closed-circuit

configuration where fluid is circulated in a loop between the hydraulic pump **104** and the actuator **306**.

(108) FIG. **12** illustrates a hydraulic system **400** having the assembly **100** operating in a closed-circuit configuration, in accordance with an example implementation. Components of the hydraulic system **400** that are similar to components of the hydraulic system **300** are designated with the same reference numbers.

(109) In the hydraulic system **400**, the hydraulic pump **104** of the assembly **100** provides fluid from the outlet port **120** through a valve assembly **402** to the first chamber **316** or the second chamber **318**, and fluid being discharged from the other chamber of the actuator **306** returns to the inlet port **118** of the assembly **100**. As such, fluid circulates between the hydraulic pump **104** of the assembly **100** and the actuator **306**.

(110) Due to the configuration of the piston **310** with the rod **314** extending through the second chamber **318**, fluid flow rate of fluid entering into or being discharged from the first chamber **316** is larger than a respective fluid flow rate of fluid entering into or being discharged from the second chamber **318**. The hydraulic system **400** includes a boost circuit **404** configured to boost the fluid flow rate, or consume any excess flow, due to a difference in fluid flow rate of fluid provided to or discharged from the first chamber **316** and fluid flow rate of fluid provided to or discharged from the second chamber **318**.

(111) The boost circuit **404** can include, for example, a charge pump that is configured to draw fluid from the fluid reservoir **302** and provide boost fluid flow to a boost flow line **406**, which is fluidly-coupled to the valve assembly **402**. The valve assembly **402** can include a valve that facilitates joining make-up fluid flow from the boost flow line **406** with fluid discharged from the second chamber **318** before providing the fluid to the inlet port **118** of the assembly **100** to increase the fluid flow rate into the inlet port **118** when the piston **310** is extending.

(112) In another example, the boost circuit **404** can comprise an accumulator configured to store pressurized fluid, and the fluid reservoir **302** might not be used. In another example, the hydraulic system includes multiple actuators and multiple assemblies similar to the assembly **100**, and the boost circuit **404** can receive excess flow from other actuators to provide it as boost flow to the actuator **306**.

(113) The boost circuit **404** can also be configured to receive excess fluid flowing through the boost flow line **406** and provide a path for such excess fluid to the fluid reservoir **302** (or to other actuators in the system). Particularly, when the piston **310** is retracting and fluid discharged from the first chamber **316** is in excess of fluid consumed required by the second chamber **318**, the excess flow is provided through the valve assembly **402** to the boost flow line **406**, then to the boost circuit **404**, which directs the fluid flow to the fluid reservoir **302** (or to other actuators in the system).

(114) Additionally, in the hydraulic system **400**, the boost circuit **404** can further be configured as the external source **500** providing cooling fluid to the assembly **100**. Particularly, the boost circuit **404** can provide cooling fluid through a cooling fluid line **408**, which is fluidly-coupled to the assembly **100** (e.g., to the fluid line **502**).

(115) In an example, the boost circuit **404** can be configured to provide pressurized fluid to the valve assembly **402** at a pressure level that is higher than the desired pressure level for the cooling fluid. For example, the boost circuit **404** may provide boost fluid flow at a pressure level of 400 psi, whereas the desired pressure level for the cooling fluid can be about 50 psi. In this case, the hydraulic system **400** can include a pressure reducing valve **410** configured to reduce the pressure level from 400 psi to 50 psi, and provide cooling fluid at a reduced pressure level downstream to the fluid line **502** providing cooling fluid to the assembly **100**.

(116) The pressure reducing valve **410** is represented symbolically in FIG. **12** and is not meant to be a limiting representation. A pilot operated pressure reducing valve could be used. Any valve or combination of components configured to reduce pressure level of fluid flowing therethrough can

be used.

(117) The cooling fluid exiting the pressure reducing valve **410** flows to the fluid line **502** then circulates through the main housing **108** (see FIGS. **7**A-**10**) and through the electronics housing cover **170** (see FIGS. **6**A-**6**B). The cooling fluid is then provided through the fluid line **526** to a cooling fluid return line, which provides the cooling fluid back to the fluid reservoir **302**. The cooling fluid return line is not shown in FIG. **12** to reduce visual clutter in the drawing. However, it should be understood that the cooling fluid discharged through the fluid line **526** can be provided directly to the fluid reservoir **302**, or indirectly through the valve assembly **402** and the boost circuit **404**.

(118) As mentioned above with respect to FIG. **11**, in examples, the electronic drive device **106** of the assembly **100** can receive sensor information indicative of the temperatures of the electric motor **102** and the inverter board **202**, and can responsively control the solenoid **514** of the flow control valve **510** to apportion cooling fluid between the electric motor **102** (e.g., to the cooling inner ring **700**, **800**, **900**, **1000**) and the electronics housing cover **170** such that the respective temperatures of the electric motor **102** and the inverter board **202** remain below safe operating temperatures.

(119) Several configuration variations can be implemented to the example implementation depicted in FIGS. **1-10**. For example, rather than having the main housing **108** coupled to the electronic device housing **140**, which is configured as a separate housing, both housings can be combined into a single housing. As another example, other types of pumps can be used. Also, in another example implementation, a gearbox might be used to couple the motor rotor **114** to the pump drive shaft **134**. Also, while the hydraulic pump **104** is disposed partially within the stator **112**, in another example implementation, the pump can be disposed completely within the windings of the stator.

(120) The detailed description above describes various features and operations of the disclosed systems with reference to the accompanying figures. The illustrative implementations described herein are not meant to be limiting. Certain aspects of the disclosed systems can be arranged and combined in a wide variety of different configurations, all of which are contemplated herein.

(121) Further, unless context suggests otherwise, the features illustrated in each of the figures may be used in combination with one another. Thus, the figures should be generally viewed as component aspects of one or more overall implementations, with the understanding that not all illustrated features are necessary for each implementation.

(122) Additionally, any enumeration of elements, blocks, or steps in this specification or the claims is for purposes of clarity. Thus, such enumeration should not be interpreted to require or imply that these elements, blocks, or steps adhere to a particular arrangement or are carried out in a particular order.

(123) Further, devices or systems may be used or configured to perform functions presented in the figures. In some instances, components of the devices and/or systems may be configured to perform the functions such that the components are actually configured and structured (with hardware and/or software) to enable such performance. In other examples, components of the devices and/or systems may be arranged to be adapted to, capable of, or suited for performing the functions, such as when operated in a specific manner.

(124) By the term “substantially” it is meant that the recited characteristic, parameter, or value need not be achieved exactly, but that deviations or variations, including for example, tolerances, measurement error, measurement accuracy limitations and other factors known to skill in the art, may occur in amounts that do not preclude the effect the characteristic was intended to provide

(125) The arrangements described herein are for purposes of example only. As such, those skilled in the art will appreciate that other arrangements and other elements (e.g., machines, interfaces, operations, orders, and groupings of operations, etc.) can be used instead, and some elements may be omitted altogether according to the desired results. Further, many of the elements that are described are functional entities that may be implemented as discrete or distributed components or

in conjunction with other components, in any suitable combination and location.

(126) While various aspects and implementations have been disclosed herein, other aspects and implementations will be apparent to those skilled in the art. The various aspects and implementations disclosed herein are for purposes of illustration and are not intended to be limiting, with the true scope being indicated by the following claims, along with the full scope of equivalents to which such claims are entitled. Also, the terminology used herein is for the purpose of describing particular implementations only, and is not intended to be limiting.

(127) Embodiments of the present disclosure can thus relate to one of the enumerated example embodiments (EEEs) listed below.

(128) EEE 1 is an assembly comprising: a main housing having an internal chamber therein; an electric motor disposed in the internal chamber of the main housing and comprising a motor rotor; a cooling inner ring disposed in the internal chamber of the main housing about the electric motor, wherein the cooling inner ring comprises one or more motor cooling fluid channels configured to receive cooling fluid from an external source of cooling fluid and allow cooling fluid to flow about the electric motor to cool the electric motor: a hydraulic pump positioned in the main housing, at least partially within the motor rotor of the electric motor; and an enclosure coupled to the main housing and comprising (i) an inverter board disposed therein, and (ii) one or more inverter cooling fluid channels configured to allow cooling fluid from the external source to cool the inverter board.

(129) EEE 2 is the assembly of EEE 1, wherein the cooling inner ring comprises a plurality of arcuate ridges such that the one or more motor cooling fluid channels are formed therebetween.

(130) EEE 3 is the assembly of any of EEEs 1-2, wherein the cooling inner ring comprises a spiral or zigzagging motor cooling fluid channel that is formed as a depression about an exterior surface of the cooling inner ring.

(131) EEE 4 is the assembly of any of EEEs 1-3, wherein the cooling inner ring comprises a plurality of protrusions, wherein respective protrusions of the plurality of protrusions have a polygonal shape, such that the one or more motor cooling fluid channels are formed between the plurality of protrusions, thereby allowing cooling fluid to be diffused therethrough.

(132) EEE 5 is the assembly of any of EEEs 1-4, wherein the main housing comprises: a motor cooling fluid inlet port configured to receive cooling fluid from the external source and provide cooling fluid to the one or more motor cooling fluid channels of the cooling inner ring; and a motor cooling fluid outlet port configured to provide cooling fluid that has flowed through the one or more motor cooling fluid channels to a fluid reservoir.

(133) EEE 6 is the assembly of any of EEEs 1-5, wherein the enclosure comprises: an electronic device housing coupled to the main housing; and an electronics housing cover coupled to the electronic device housing, such that the enclosure is formed by the electronics housing cover and the electronic device housing.

(134) EEE 7 is the assembly of EEE 6, wherein the inverter board is mounted to the electronics housing cover, and wherein the one or more inverter cooling fluid channels are formed in the electronics housing cover.

(135) EEE 8 is the assembly of EEE 7, further comprising: a back plate coupled to the electronics housing cover, wherein the back plate comprises: (i) an inverter cooling fluid inlet port configured to receive cooling fluid from the external source and provide cooling fluid to the one or more inverter cooling fluid channels formed in the electronics housing cover, and (ii) an inverter cooling fluid outlet port configured to receive cooling fluid that has flowed through the one or more inverter cooling fluid channels and provide cooling fluid to a fluid reservoir.

(136) EEE 9 is the assembly of EEE 8, wherein a portion of an inverter cooling fluid channel of the one or more inverter cooling fluid channels is formed in the electronics housing cover and another mating portion of the inverter cooling fluid channel is formed in the back plate.

(137) EEE 10 is the assembly of any of EEEs 1-9, further comprising: a flow control valve configured to apportion cooling fluid between the cooling inner ring and the enclosure.

(138) EEE 11 is a hydraulic system comprising: an actuator having a first chamber and a second chamber: a valve assembly configured to control fluid flow to and from the first chamber and the second chamber of the actuator; and an assembly comprising: a main housing having an internal chamber therein, an electric motor disposed in the internal chamber of the main housing and comprising a motor rotor, a cooling inner ring disposed in the internal chamber of the main housing about the electric motor, wherein the cooling inner ring comprises one or more motor cooling fluid channels configured to receive cooling fluid from an external source of cooling fluid and allow cooling fluid to flow about the electric motor to cool the electric motor, wherein the main housing comprises: (i) a motor cooling fluid inlet port configured to receive cooling fluid and provide cooling fluid to the one or more motor cooling fluid channels, and (ii) a motor cooling fluid outlet port configured to discharge cooling fluid that has flowed through the one or more motor cooling fluid channels, a hydraulic pump positioned in the main housing, at least partially within the motor rotor of the electric motor, wherein the hydraulic pump is configured to receive fluid from an inlet port and provide fluid flow to an outlet port that is fluidly-coupled to the valve assembly, such that the hydraulic pump is configured to provide fluid flow to the actuator via the valve assembly, and an enclosure coupled to the main housing and comprising (i) an inverter board disposed therein, and (ii) one or more inverter cooling fluid channels configured to allow cooling fluid from the external source to cool the inverter board, (iii) an inverter cooling fluid inlet port configured to receive cooling fluid and provide cooling fluid to the one or more inverter cooling fluid channels, and (iv) an inverter cooling fluid outlet port configured to discharge cooling fluid that has flowed through the one or more inverter cooling fluid channels.
(139) EEE 12 is the hydraulic system of EEE 11, further comprising: a fluid reservoir, wherein the inlet port of the hydraulic pump is fluidly-coupled to the fluid reservoir such that the hydraulic pump draws fluid from the fluid reservoir, and wherein the valve assembly is fluidly-coupled to the fluid reservoir to provide fluid discharged from the actuator to the fluid reservoir, wherein a portion of fluid discharged from the actuator branches out to be provided to the motor cooling fluid inlet port and the inverter cooling fluid inlet port, such that the actuator is the external source of cooling fluid.
(140) EEE 13 is the hydraulic system of EEE 12, further comprising: a valve disposed in a fluid line that fluidly couples the actuator and the fluid reservoir, wherein the valve is configured to increase pressure level upstream of the valve, and wherein the portion of fluid provided to the motor cooling fluid inlet port and the inverter cooling fluid inlet port is branched out at a junction disposed upstream of the valve.
(141) EEE 14 is the hydraulic system of any of EEEs 11-13, wherein the assembly is configured to (i) receive fluid discharged from the actuator through the valve assembly at the inlet port, and (ii) provide fluid via the outlet port through the valve assembly to the actuator, wherein the hydraulic system further comprises: a boost circuit that is fluidly-coupled to the valve assembly, wherein the boost circuit is configured to provide boost fluid flow and receive excess fluid flow due to a difference in fluid flow rate of fluid provided to or discharged from the first chamber and fluid flow rate of fluid provided to or discharged from the second chamber, and wherein the boost circuit is fluidly-coupled to the motor cooling fluid inlet port and the inverter cooling fluid inlet port to provide cooling fluid thereto, such that the boost circuit is the external source of cooling fluid.
(142) EEE 15 is the hydraulic system of EEE 14, further comprising: a pressure reducing valve configured to receive cooling fluid from the boost circuit and provide cooling fluid at a reduced pressure level to the motor cooling fluid inlet port and the inverter cooling fluid inlet port.
(143) EEE 16 is the hydraulic system of any of EEEs 11-15, wherein the enclosure comprises: an electronic device housing coupled to the main housing; and an electronics housing cover coupled to the electronic device housing, such that the enclosure is formed by the electronics housing cover and the electronic device housing.
(144) EEE 17 is the hydraulic system of EEE 16, wherein the inverter board is mounted to the

electronics housing cover, and wherein the one or more inverter cooling fluid channels are formed in the electronics housing cover.
(145) EEE 18 is the hydraulic system of EEE 17, wherein the assembly further comprises: a back plate coupled to the electronics housing cover, wherein the back plate comprises the inverter cooling fluid inlet port and the inverter cooling fluid outlet port, and wherein a portion of an inverter cooling fluid channel of the one or more inverter cooling fluid channels is formed in the electronics housing cover and another mating portion of the inverter cooling fluid channel is formed in the back plate.
(146) EEE 19 is the hydraulic system of any of EEEs 11-18, further comprising: a flow control valve configured to apportion cooling fluid between the cooling inner ring and the enclosure.
(147) EEE 20 is the hydraulic system of any of EEE 11-19, wherein the cooling inner ring comprises: a plurality of arcuate ridges such that the one or more motor cooling fluid channels are formed therebetween, a spiral motor cooling fluid channel that is formed as a looping depression about an exterior surface of the cooling inner ring, a zigzagging motor cooling fluid channel that is formed as a depression about an exterior surface of the cooling inner ring, or a plurality of protrusions, wherein respective protrusions of the plurality of protrusions have a polygonal shape, such that the one or more motor cooling fluid channels are formed between the plurality of protrusions, thereby allowing cooling fluid to be diffused therethrough.

Claims

1. An assembly comprising: a main housing having an internal chamber therein; an electric motor disposed in the internal chamber of the main housing and comprising a motor rotor; a cooling inner ring disposed in the internal chamber of the main housing about the electric motor, wherein the cooling inner ring comprises one or more motor cooling fluid channels configured to receive cooling fluid from an external source of cooling fluid and allow cooling fluid to flow about the electric motor to cool the electric motor; a hydraulic pump positioned in the main housing, at least partially within the motor rotor of the electric motor such that the hydraulic pump is driven by the motor rotor; and an enclosure coupled to the main housing and comprising (i) an inverter board disposed therein, and (ii) one or more inverter cooling fluid channels configured to allow cooling fluid from the external source to cool the inverter board.
2. The assembly of claim 1, wherein the cooling inner ring comprises a plurality of arcuate ridges such that the one or more motor cooling fluid channels are formed therebetween.
3. The assembly of claim 1, wherein the cooling inner ring comprises a spiral or zigzagging motor cooling fluid channel that is formed as a depression about an exterior surface of the cooling inner ring.
4. The assembly of claim 1, wherein the cooling inner ring comprises a plurality of protrusions, wherein respective protrusions of the plurality of protrusions have a polygonal shape, such that the one or more motor cooling fluid channels are formed between the plurality of protrusions, thereby allowing cooling fluid to be diffused therethrough.
5. The assembly of claim 1, wherein the main housing comprises: a motor cooling fluid inlet port configured to receive cooling fluid from the external source and provide cooling fluid to the one or more motor cooling fluid channels of the cooling inner ring; and a motor cooling fluid outlet port configured to provide cooling fluid that has flowed through the one or more motor cooling fluid channels to a fluid reservoir.
6. The assembly of claim 1, wherein the enclosure comprises: an electronic device housing coupled to the main housing; and an electronics housing cover coupled to the electronic device housing, such that the enclosure is formed by the electronics housing cover and the electronic device housing.
7. The assembly of claim 6, wherein the inverter board is mounted to the electronics housing cover,

and wherein the one or more inverter cooling fluid channels are formed in the electronics housing cover.
8. The assembly of claim 7, further comprising: a back plate coupled to the electronics housing cover, wherein the back plate comprises: (i) an inverter cooling fluid inlet port configured to receive cooling fluid from the external source and provide cooling fluid to the one or more inverter cooling fluid channels formed in the electronics housing cover, and (ii) an inverter cooling fluid outlet port configured to receive cooling fluid that has flowed through the one or more inverter cooling fluid channels and provide cooling fluid to a fluid reservoir.
9. The assembly of claim 8, wherein a portion of an inverter cooling fluid channel of the one or more inverter cooling fluid channels is formed in the electronics housing cover and another mating portion of the inverter cooling fluid channel is formed in the back plate.
10. The assembly of claim 1, further comprising: a flow control valve configured to apportion cooling fluid between the cooling inner ring and the enclosure.
11. A hydraulic system comprising: an actuator having a first chamber and a second chamber; a valve assembly configured to control fluid flow to and from the first chamber and the second chamber of the actuator; and an assembly comprising: a main housing having an internal chamber therein, an electric motor disposed in the internal chamber of the main housing and comprising a motor rotor, a cooling inner ring disposed in the internal chamber of the main housing about the electric motor, wherein the cooling inner ring comprises one or more motor cooling fluid channels configured to receive cooling fluid from an external source of cooling fluid and allow cooling fluid to flow about the electric motor to cool the electric motor, wherein the main housing comprises: (i) a motor cooling fluid inlet port configured to receive cooling fluid and provide cooling fluid to the one or more motor cooling fluid channels, and (ii) a motor cooling fluid outlet port configured to discharge cooling fluid that has flowed through the one or more motor cooling fluid channels, a hydraulic pump positioned in the main housing, at least partially within the motor rotor of the electric motor, wherein the hydraulic pump is configured to receive fluid from an inlet port and provide fluid flow to an outlet port that is fluidly-coupled to the valve assembly, such that the hydraulic pump is configured to provide fluid flow to the actuator via the valve assembly, and an enclosure coupled to the main housing and comprising (i) an inverter board disposed therein, and (ii) one or more inverter cooling fluid channels configured to allow cooling fluid from the external source to cool the inverter board, (iii) an inverter cooling fluid inlet port configured to receive cooling fluid and provide cooling fluid to the one or more inverter cooling fluid channels, and (iv) an inverter cooling fluid outlet port configured to discharge cooling fluid that has flowed through the one or more inverter cooling fluid channels.
12. The hydraulic system of claim 11, further comprising: a fluid reservoir, wherein the inlet port of the hydraulic pump is fluidly-coupled to the fluid reservoir such that the hydraulic pump draws fluid from the fluid reservoir, and wherein the valve assembly is fluidly-coupled to the fluid reservoir to provide fluid discharged from the actuator to the fluid reservoir, wherein a portion of fluid discharged from the actuator branches out to be provided to the motor cooling fluid inlet port and the inverter cooling fluid inlet port, such that the actuator is the external source of cooling fluid.
13. The hydraulic system of claim 12, further comprising: a valve disposed in a fluid line that fluidly couples the actuator and the fluid reservoir, wherein the valve is configured to increase pressure level upstream of the valve, and wherein the portion of fluid provided to the motor cooling fluid inlet port and the inverter cooling fluid inlet port is branched out at a junction disposed upstream of the valve.
14. The hydraulic system of claim 11, wherein the assembly is configured to (i) receive fluid discharged from the actuator through the valve assembly at the inlet port, and (ii) provide fluid via the outlet port through the valve assembly to the actuator, wherein the hydraulic system further comprises: a boost circuit that is fluidly-coupled to the valve assembly, wherein the boost circuit is

configured to provide boost fluid flow and receive excess fluid flow due to a difference in fluid flow rate of fluid provided to or discharged from the first chamber and fluid flow rate of fluid provided to or discharged from the second chamber, and wherein the boost circuit is fluidly-coupled to the motor cooling fluid inlet port and the inverter cooling fluid inlet port to provide cooling fluid thereto, such that the boost circuit is the external source of cooling fluid.
15. The hydraulic system of claim 14, further comprising: a pressure reducing valve configured to receive cooling fluid from the boost circuit and provide cooling fluid at a reduced pressure level to the motor cooling fluid inlet port and the inverter cooling fluid inlet port.
16. The hydraulic system of claim 11, wherein the enclosure comprises: an electronic device housing coupled to the main housing; and an electronics housing cover coupled to the electronic device housing, such that the enclosure is formed by the electronics housing cover and the electronic device housing.
17. The hydraulic system of claim 16, wherein the inverter board is mounted to the electronics housing cover, and wherein the one or more inverter cooling fluid channels are formed in the electronics housing cover.
18. The hydraulic system of claim 17, wherein the assembly further comprises: a back plate coupled to the electronics housing cover, wherein the back plate comprises the inverter cooling fluid inlet port and the inverter cooling fluid outlet port, and wherein a portion of an inverter cooling fluid channel of the one or more inverter cooling fluid channels is formed in the electronics housing cover and another mating portion of the inverter cooling fluid channel is formed in the back plate.
19. The hydraulic system of claim 11, further comprising: a flow control valve configured to apportion cooling fluid between the cooling inner ring and the enclosure.
20. The hydraulic system of claim 11, wherein the cooling inner ring comprises: a plurality of arcuate ridges such that the one or more motor cooling fluid channels are formed therebetween, a spiral motor cooling fluid channel that is formed as a looping depression about an exterior surface of the cooling inner ring, a zigzagging motor cooling fluid channel that is formed as a depression about an exterior surface of the cooling inner ring, or a plurality of protrusions, wherein respective protrusions of the plurality of protrusions have a polygonal shape, such that the one or more motor cooling fluid channels are formed between the plurality of protrusions, thereby allowing cooling fluid to be diffused therethrough.